

Demonstrating VRScroll: A Shape-Changing Device for Precise Sketching in Virtual Reality

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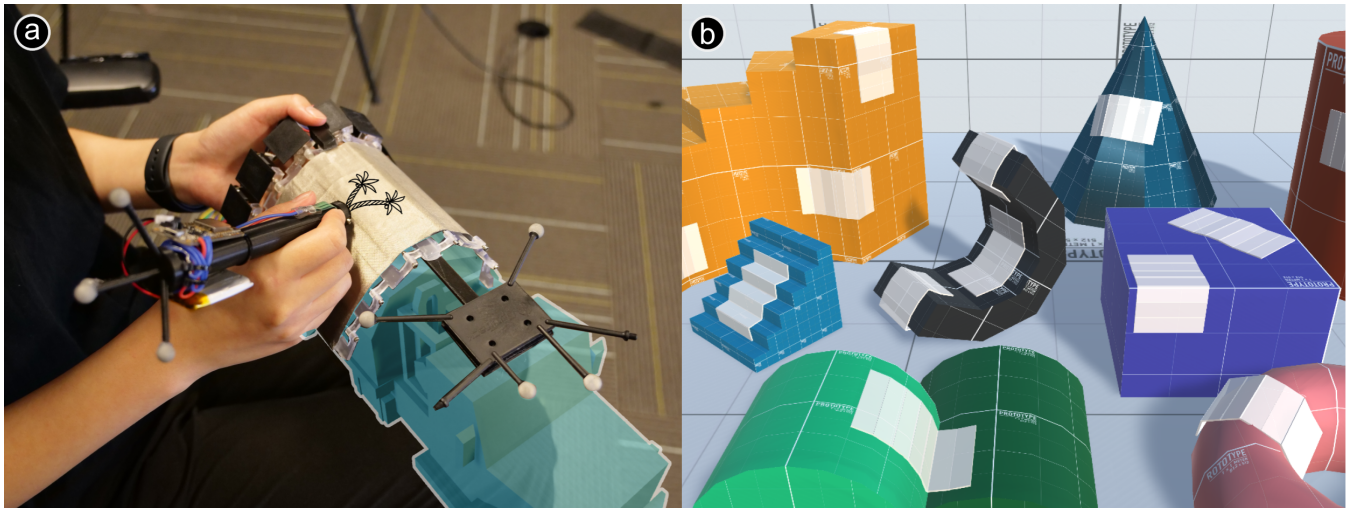


Figure 1: *VRScroll* is a dynamic shape display developed to enhance sketching in VR by physically approximating the shape of the virtual surface. (a) *VRScroll* physically rendering the curved surface of a tank truck. (b) *VRScroll* can mimic the surface of various virtual shapes.

ABSTRACT

Sketching precisely in Virtual Reality (VR) is challenging due to the lack of a physical surface for users to lean on, feel the contact and the movement of the pen, and constrain the pen movement. Using physical surface interfaces, such as a touch tablet, can improve sketching performance; however, they cannot replicate the diverse shapes of virtual objects. We present *VRScroll*, a novel shape-changing device equipped with a dynamic drawing surface to support high-precision sketching in VR. The device has seven motor-controlled flaps that can independently change the angles between them in real time to mimic the surface shape of a virtual object. Since the degree of freedom and resolution of *VRScroll* is limited, it cannot replicate the surface geometry of diverse types of virtual objects. To allow the device to closely mimic the shape of the virtual surface, we developed a shape approximation algorithm.

CCS CONCEPTS

• Human-centered computing → Haptic devices.

KEYWORDS

sketching, dynamic shape display, shape approximation, virtual reality

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1 INTRODUCTION

Virtual and mixed reality environments provide designers with an extensive and immersive 3D workspace that facilitates intuitive and unimpeded manipulation of 3D models [14]. While designers can utilize the 6-DOF interaction and spatial representation offered by VR technology, routine design tasks such as precise sketching and annotating on models remain a significant challenge due to the absence of a physical drawing board for guiding strokes and resting users' hands [13].

To improve the precision and ergonomics of VR sketching, previous research has incorporated sketching tablets in mixed reality to

allow users to draw on virtual objects with passive haptic feedback, feel of contact and constrained pen movement provided by the planar physical surface [1, 6]. While using sketching tablets can improve performance, it has challenges with drawing on non-planar surfaces due to the lost geometry information and the discrepancy between virtual and physical surfaces. A prior study showed that it may result in VR sketching being unintuitive, technically challenging, and learning-demanding [19]. On the other hand, fabricating tangible proxies of virtual objects can be helpful for designers to make high-precision drawings [28]. However, building these proxies is both a time-consuming and resource-intensive process, and it may be impractical if the designer needs to work with a variety of virtual objects in different shapes.

In this work, we propose to use a dynamic shape display to create a sketching surface that mimics the surface of a virtual object so that the designer can achieve precise sketching on both planar and non-planar surfaces. To facilitate this idea, we developed *VRScroll*, a novel dynamic shape-changing interface with a sketching surface (Figure 1b). *VRScroll* consists of seven 3D-printed flaps. Each flap is equipped with a DC motor and an encoder so that the angle between the flaps can be independently and accurately controlled to mimic the surface shape of a virtual object. Due to the limited resolution (7 flaps) and one-dimensional shape controllability, *VRScroll* can only accurately replicate a limited range of shapes, such as planes, folded planes, zigzags, and sharp corners. To accommodate a more diverse range of shapes, we developed a shape approximation method for *VRScroll*. This method determines the flap configuration of *VRScroll* that closely matches the contours of the original surface (Figure 1a).

We demonstrate the use of *VRScroll* with shape approximation in our sketch application. In the sketch application, the user can snap *VRScroll* to a virtual object's surface. This will transform the shape of *VRScroll* to mimic the shape of the virtual surface so that the user can sketch on it using a force-sensitive sketching pen as if they are sketching directly on the virtual object (Figure 1b).

2 RELATED WORK

2.1 Physical Surfaces for VR Sketching

Physical surfaces such as touchscreens and graphic tablets have also been utilized in many pen-based design systems in mixed reality for detailed sketching [1, 6, 7] or model manipulation [5, 25, 27]. However, planar physical surfaces may not be sufficient to support direct and intuitive interaction for the variety of surfaces in VR. While these design systems have the capability to project a non-planar virtual surface onto a flat physical surface and offer a certain degree of haptic feedback to designers, critical model information pertaining to physics and geometry remains absent. There can be a discrepancy between the virtual and physical surfaces, which may impact the user's interaction performance and immersion. Additionally, sketching 3D models on a 2D surface can be unintuitive because it necessitates reflections of spatiality and proportionateness [19], and the transfer of 2D sketches into 3D models is technically challenging and learning-demanding [1, 3].

To enable direct sketching or writing on a virtual model, researchers have investigated the fabrication of the one-to-one tangible proxy and the alignment between virtual and physical objects [23]. The precise geometric information of tangible proxies

can provide physical guidance in VR, resulting in a significantly lower line-based sketching deviation [28]. However, fabricating an exact physical replica of each virtual object can be expensive, time-consuming, and impractical due to the diversity of their shapes [10, 11, 20, 24]. Even if the user has fabricated plenty of proxies for interaction, switching between physical objects may lead to increased task time and mode errors [21].

2.2 Dynamic Shape Displays

Dynamic shape displays have been extensively explored in VR to simulate various virtual objects and display tangible surfaces with different properties such as size, modularity, form, curvature, closure, and texture [15, 15, 18]. Some shape displays utilize an array of actuated pins to render high-resolution shapes so that users can easily recognize different virtual objects by touching [4, 8, 22]. Other actuators such as motors and mini robots have also been investigated to provide shape information of virtual objects [26, 29]. While these novel shape displays show great potential in facilitating user perception of virtual objects, including touching, tracing, and even breaking objects, they lack a sufficiently large and smooth surface for designers to interact with 3D models by sketching or writing on their surfaces.

Bendable shape displays possess a relatively smoother and larger surface that is constructed by flaps [9, 16, 20]. These displays can be manipulated by rotating the flaps with motors to change their shapes. However, current bendable shape displays predominantly focus on simulating objects that function as grasping tools to create a tangible and immersive experience in VR, rather than providing an interaction surface to support precise and high-performance interactions such as sketching. Morphaces might be the closest to our work and demonstrates the potential use of a deformable surface for sketching [17]. However, the study only explores the manual deformation of physical surfaces and does not investigate how the system creates deformed surfaces that match the surfaces of virtual objects.

3 DYNAMIC SHAPE-CHANGING INTERFACE FOR PRECISE SKETCHING IN VR

3.1 VRScroll: A Dynamic Shape Display

VRScroll is a dynamic shape display designed to help users precisely sketch in VR by changing its surface to mimic the surface of the virtual object (Figure 2). Bending-deformation techniques have been extensively explored to create dynamic surfaces, and *VRScroll* features 7 flaps to support its shape deformation. We chose a motor-driven bending mechanism [16] for changing the shape because of its extendability, quick and accurate response, and ease of development.

VRScroll has a dimension of $190\text{mm} \times 120\text{mm} \times 4\text{mm}$ and weighs 157 g. Each flap in *VRScroll* is fabricated with resin material using an SLA 3D printer, and the rotation of each flap is controlled by a PID controller using a DC geared motor (Adafruit N20, 100 rpm, 1:100 gear ratio) and an encoder. We used three dual motor drivers (Adafruit TB6612) to drive the six motors, and two Arduino UNO WiFi REV 2 microcontrollers were used to control the entire system. All motor drivers and microcontrollers were powered by a 2S Li-Po

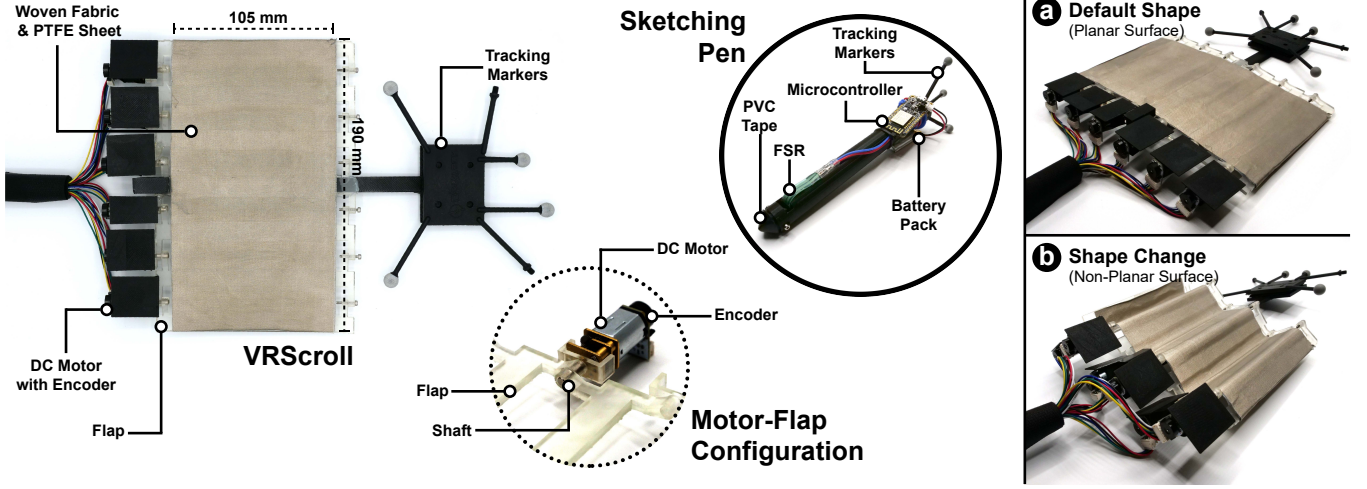


Figure 2: VRScroll device. Example surfaces created by VRScroll: (a) planar surface, (b) non-planar surface.

battery (7.4V, 450mAh). Each flap supports a rotation range of ~ 280 degrees, allowing VRScroll to mimic a wide range of shapes.

A $190\text{mm} \times 105\text{mm}$ flexible PTFE sheet and a woven fabric cover the flaps to create a continuous and smooth touchable surface. The number of flaps on the device can be modified. However, we chose a 7 flap configuration to minimize the fabrication workload and construct a device that is comparatively lightweight, yet able to fulfill the size requirements of a sketching pad and offer a sufficient array of shapes for our designated sketching tasks. Finally, reflective markers are attached to VRScroll for it to be tracked in 6-DOF by the OptiTrack tracking system.

3.2 Sketching Pen

We implemented a sketching pen for users to create sketches on the VRScroll (Figure 2). The sketching pen is designed to resemble the shape of a marker pen and has a height of 170 mm, a radius of 8.5 mm, and weighs 47 g. We placed a force-sensitive resistor (FSR) between the tip and the barrel of the pen to detect the contact between the surface and the pen. To enhance the smoothness of contact, we wrapped black PVC tape around the pen tip. The pen is equipped with an Adafruit Feather HUZZAH ESP8266 board powered by a 3.7 v Li-Po battery (900mAh) for measuring the force and transmitting it to the PC running the sketch application using UDP via Wi-Fi. Similar to VRScroll, the sketching pen also has reflective markers attached to it for optical tracking.

3.3 Shape Approximation

Shape approximation techniques have been thoroughly explored in prior literature [12]. However, most of these techniques require prior knowledge of the original shape's geometry, which may be unavailable for objects encountered dynamically in a VR environment. We avoid this by using a collision-based approach for dynamically approximating the shapes of virtual objects.

Our shape approximation algorithm features a closed-loop feedback system following three steps: 1) align the direction of the

VRScroll with the virtual object, 2) find the anchor to place the virtual instance of VRScroll on the virtual object, 3) iteratively rotate each flap of VRScroll until a collision is detected between the flap and the virtual surface of the target object, and eventually wraps VRScroll around part of the virtual object and snaps on its surface.

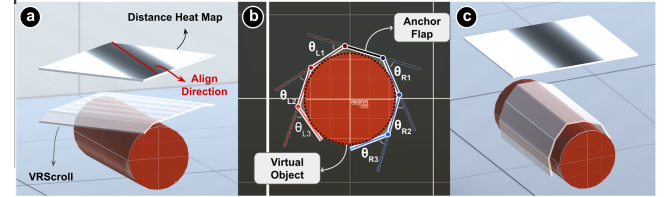


Figure 3: Shape Approximation Algorithm. (a) Align VRScroll with the virtual object by analyzing a distance heat map. (b) Estimate the flap angles by locating the anchor flap and adjusting the side flaps while performing collision detection. (c) End the shape approximation when all flaps collide with the virtual object's surface.

Since the shape change of VRScroll is 1-dimensional, VRScroll may wrap the virtual object surface most accurately when the bending direction of the device is aligned with the direction of maximum curvature of the virtual object (Figure 3a). To compute this, $8 \times 32 \times 7$ (flaps) points are sampled from VRScroll to render a 56×32 pixel distance heat map based on the normalized distance from sampling points and the virtual object. We then use the principal component analysis to find the aligned direction with the largest variance on this heat map, which shows the direction of the virtual object. VRScroll is then rotated to be parallel with this direction, so that the direction of shape change is perpendicular to the aligned direction.

In the wrapping algorithm, the 7-flap structure of VRScroll can be divided into three sections – anchor (middle flap in Figure 3a), anchor's left, and anchor's right section. The anchor of VRScroll contains only one flap that is closest to the target virtual object,

which determines the root position and orientation of the virtual VRScroll. The remaining six flaps need to be adjusted individually, and in an inside-out order starting from the anchor flap as illustrated in Figure 3b. For each flap, if the flap initially collides with the target, it will be rotated to decrease the distance from the flap to the target until the collision disappears or the distance is less than 5 mm. If the flap initially does not collide with the target, it will be rotated to decrease the distance from the flap to the target until it collides with the target or the distance is less than 5 mm.

We implemented this algorithm in Unity, where the intersection between each flap's and the virtual object's Collider was used to determine a collision. The shape of a virtual surface is considered to be successfully approximated if all the flaps are able to sequentially find their points of collision (Figure 3c), upon which the rotation value of each virtual flap is extracted and passed to the motors of the actual VRScroll device to mimic the approximated shape.

4 SKETCH APPLICATION IN VR

To demonstrate the use of VRScroll with shape approximation, we developed a sketch application for users to manipulate, draw, and write on various virtual objects in VR.

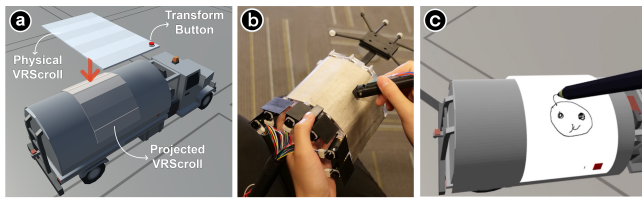


Figure 4: VRScroll Interaction. (a) VRScroll is projected over a virtual surface to visualize the shape approximation. (b) VRScroll's shape mimics the projected VRScroll. (c) Sketches are rendered directly on the virtual surface.

The VRScroll interaction for sketching in VR involves three steps - *projection*, *transformation*, and *sketching*. First, the system shows a *projection* of VRScroll on the virtual surface placed beneath it. The user can move the VRScroll device to adjust its projected position on the virtual surface where they intend to sketch, and visually inspect the shape-approximation that can be afforded by VRScroll (Figure 4a). Once the user is satisfied with the positioning of VRScroll and the shape approximation, they can press the *transform* button on the virtual VRScroll (Figure 4a) to change the shape of VRScroll to mimic the virtual surface (Figure 4b). Simultaneously, the target virtual surface is snapped to VRScroll, allowing the user to start sketching on the surface using the physical VRScroll. Finally, the user can use the pen to create *sketches* on the physical VRScroll, which are tracked in real-time and simultaneously rendered on the virtual surface (Figure 4c).

5 CONCLUSION

Despite the advantages of utilizing physical surfaces for sketching in VR [2], having a physical surface that mimics the shape of a virtual surface remains a challenge [28]. This work aims to address the challenge by providing a physical surface that mimics the shape of various virtual surfaces. While it has a limited number of flaps and

cannot accurately replicate all shapes, it leverages a shape approximation algorithm to minimize the discrepancy between the virtual and physical surfaces, enhancing the user's sketch performance and providing a more realistic VR sketching experience.

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